

Task: Maths Behind Simple Linear Regression

classmate

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With Gradient Descent

37 ~~Greedy algo~~

17 ~~Can't learn huge and noisy~~

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

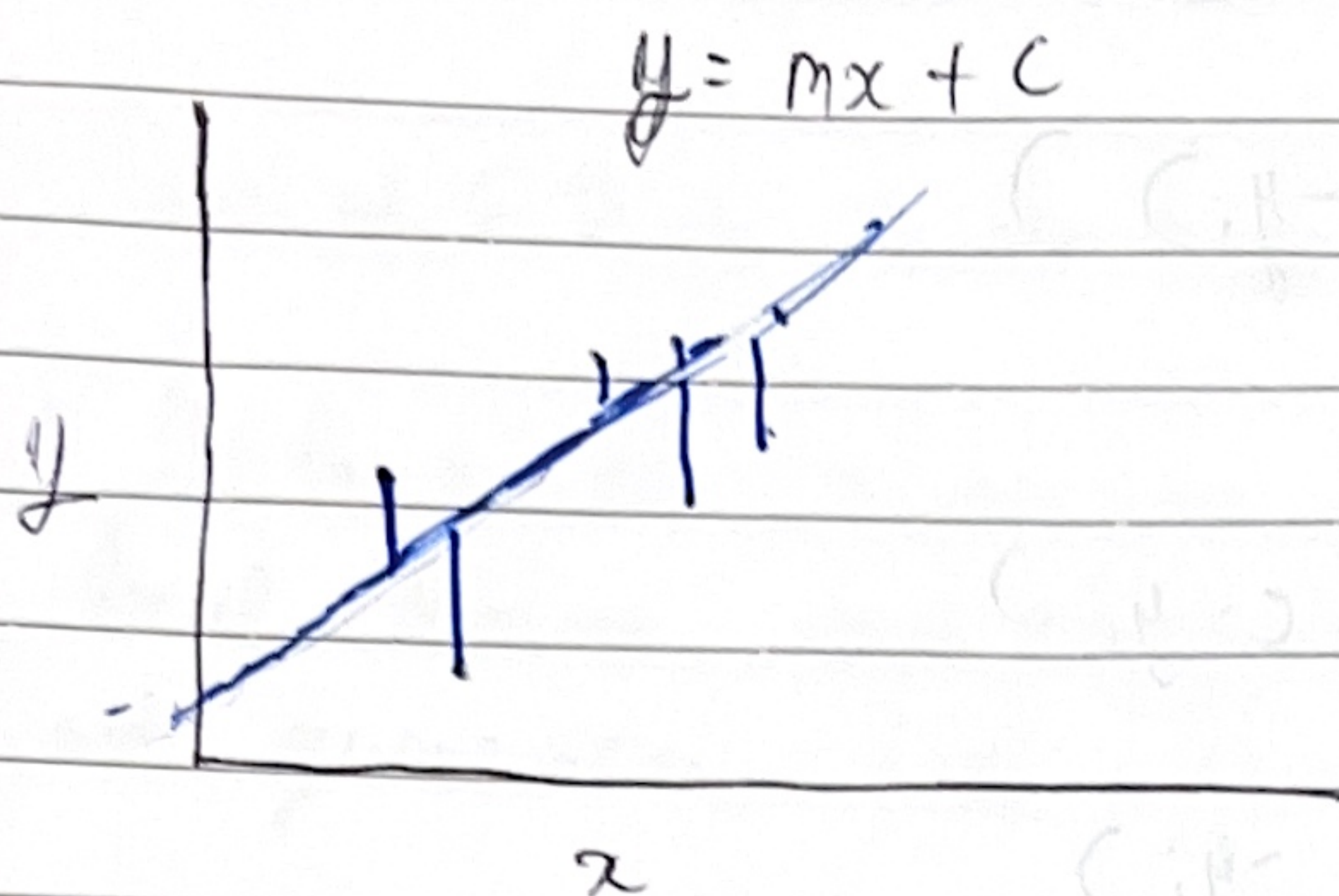
↓
Loss

func

Actual value Predicted value

$$SE = \sum (y_i - \hat{y}_i)^2$$

$$E = \sum (y_i - \hat{y}_i)$$



In ~~Greedy~~ G.D we
minimize the
MSE $\Rightarrow$ value of Loss
func

$\rightarrow$ To perform that we need to have specific value of m and c such that Loss func is minimum.

$$m_{\text{new}} = m_{\text{old}} - \eta * \frac{dL}{dm}$$

$\eta =$ Learning rate

gradient wrt m

$\eta =$

$L =$ Loss func

$$c_{\text{new}} = c_{\text{old}} - \eta * \frac{dL}{dc}$$

gradient wrt c

$$L(m, c) = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Value of Loss function depends on m and c

hence we need to change m and c to minimize Loss func

$$\Rightarrow L(m, c) = \frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2 \quad (\hat{y}_i = mx + c)$$

$$= \frac{1}{n} \sum (mx + c - y_i)^2$$

find $\frac{dL}{dm}$

$$\circ \text{ find } \frac{dL}{dm} = \frac{2}{n} \sum (mx + c - y_i) \cdot x$$

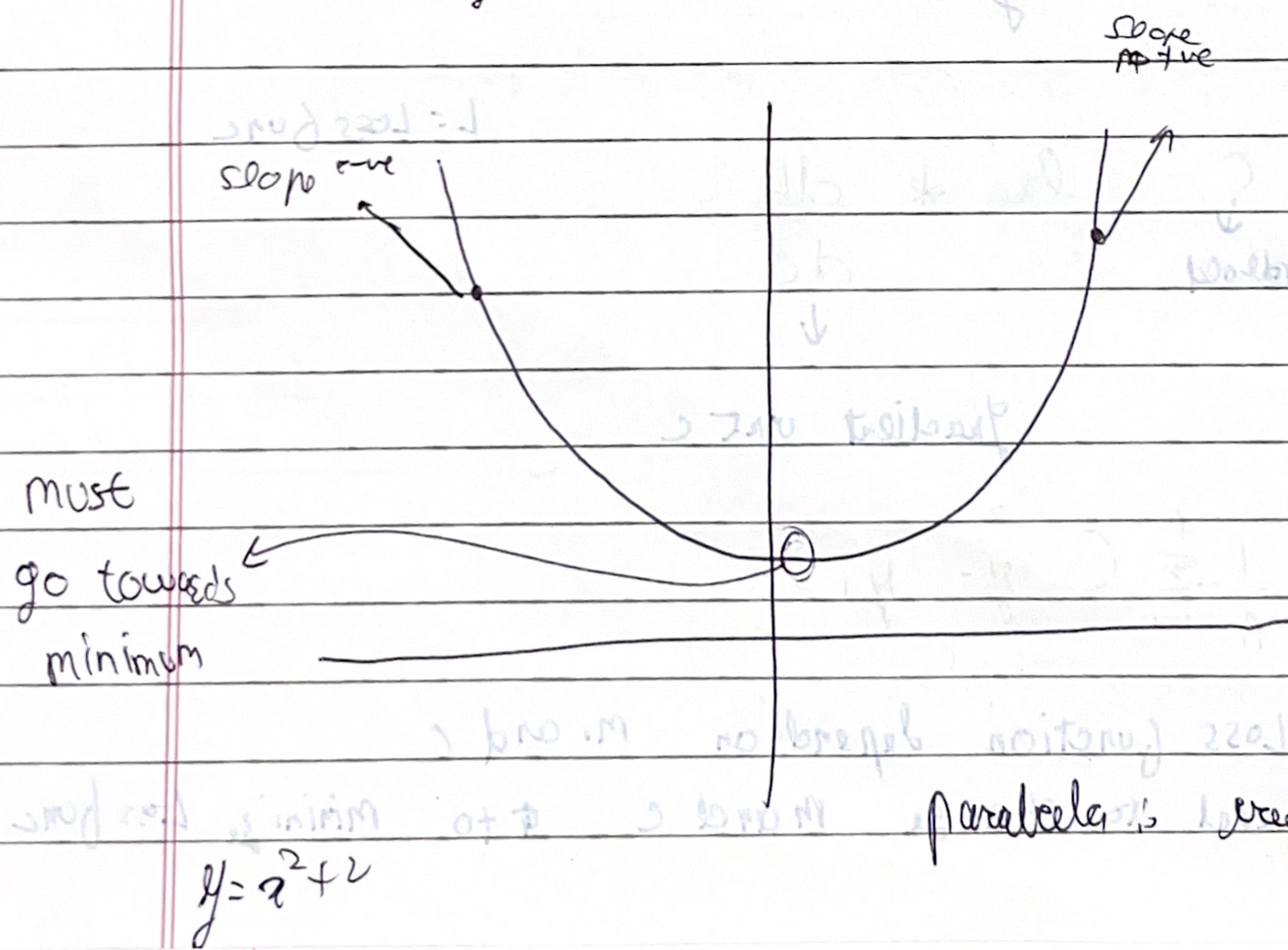
$$\Rightarrow \frac{2}{n} \sum (x * (\hat{y}_i - y_i))$$

$$\circ \text{ find } \frac{dL}{dc} = \frac{2}{n} \sum (mx + c - y_i)$$

$$= \frac{2}{n} \sum (\hat{y}_i - y_i)$$

→ The gradient descent works in the following way

- a) if value of m/c is +ve then its value will dec
 b) if value of m/c is -ve then its value will inc



- > The parabola we discuss here is a convex function
(only one)
- > Convex function: Any local minima is a global minima
Can have multiple global minima points.
- > Non Convex function: Can have multiple local minima
But has one or more global minima point
- > Plateau: A region where gradient is close to 0.
- > Goal of G.D is to minimize loss func, to reach global minima
- > With a non-convex func, GD can converge to a local minima and simply increasing no of iterations does not guarantee that it will reach global minima